

#python task 6

#list tasks

""#1.create a list of 5 student names

```
a=["a","v","d","y","g"]
```

```
print(a)""
```

""#2.add a new name to the list

```
a=["a","v","d","y","g"]
```

```
z=a.append("f")
```

```
print(z)""
```

""#3.remove a student name from the list

```
a=["a","v","d","y","g"]
```

```
z=a.remove("y")
```

```
print(z)""
```

""#4.print all elements using loop

```
a=["a","v","d","y","g"]
```

```
for z in a:
```

```
    print(z)""
```

""#5.find largest and shortest number in list

```
a=[45,9,88,56,97]
```

```
b=max(a)
```

```
c=min(a)
```

```
print(b)
```

```
print(c)"""
```

```
"""#6.count even and odd numbers in list
```

```
a=[45,9,88,56,97]
```

```
even=0
```

```
odd=0
```

```
for c in a:
```

```
    if a%2==0:
```

```
        even +=1
```

```
        print("even=",even)
```

```
    else:
```

```
        odd +=1
```

```
        print("odd=",odd)"""
```

```
"""#7.reverse a list
```

```
a=[45,9,88,56,97]
```

```
z=a.reverse()
```

```
print(z)"""
```

""#8.sort a list in ascending order

```
a=[45,9,88,56,97]
```

```
z=a.sort()
```

```
print(z)""
```

""#9.find sum and average of list elements

```
a=[45,9,88,56,97]
```

```
total=sum(a)
```

```
avg=total/len(a)
```

```
print("sum=",total)
```

```
print("average=",avg)""
```

""#10.search an element in a list

```
a=[45,9,88,56,97]
```

```
x=88
```

```
if x in a:
```

```
    print("element found")
```

```
else:
```

```
    print("element not found")""
```

#tuple tasks

""#1.create a tuple with 5 numbers

```
t=(45,9,88,56,97)
```

```
print(a)"""
```

```
"""#2.access tuple elements using indeing
```

```
t=(45,9,88,56,97)
```

```
a=t[0]
```

```
b=t[4]
```

```
print(a)
```

```
print(b)"""
```

```
"""#3.find length of tuple
```

```
t=(45,9,88,56,97)
```

```
a=len(t)
```

```
print(a)"""
```

```
"""#4.count occurence of an element
```

```
t=(45,9,88,56,97,45)
```

```
a=t.count(45)
```

```
print(a)"""
```

```
"""#5.find index position of element
```

```
t=(45,9,88,56,97)
```

```
a=t.index(45)
```

```
print(a)'''
```

```
'''#6.slice a tuple
```

```
t=(45,9,88,56,97,45)
```

```
z=t[1:4]
```

```
print(z)'''
```

```
'''#7.convert tuple into list
```

```
t=(45,9,88,56,97,45)
```

```
z=list(t)
```

```
print(z)'''
```

```
'''#8.convert list into tuple
```

```
t=[45,9,88,56,97,45]
```

```
z=tuple(t)
```

```
print(z)'''
```

```
'''#9.find maximum and minimum value
```

```
t=[45,9,88,56,97,45]
```

```
a=max(t)
```

```
b=min(t)
```

```
print(a)
```

```
print(b)"""
```

```
"""#10.iterate tuple using loop
```

```
t=[45,9,88,56,97,45]
```

```
for i in t:
```

```
    print(i)"""
```

```
#set tasks
```

```
"""#1.create a set of numbers
```

```
t={45,9,88,56,97,45}
```

```
print(t)"""
```

```
"""#2.add elements into set
```

```
t={45,9,88,56,97,45}
```

```
a=t.add(34)
```

```
print(t)"""
```

```
"""#3.remove elements from set
```

```
t={45,9,88,56,97,45}
```

```
a=t.remove(45)
```

```
print(t)"""
```

""#4.find union of two sets

t={45,9,88,56,97,45}

a={67,45}

b=a.union(t)

print(b)""

""#5.FIND INTERSECTION of two sets

a={4,5,9,8,7}

b={8,2,5,6,7,9,5,6}

print(a.intersection(b))""

""#6.find difference between sets

a={4,5,9,8,7}

b={8,2,5,6,7,9,5,6}

print(a.difference(b))""

""#7.remove duplicates from list using set

a=[4,5,9,8,7]

b=list(set(a))

print(b)""

""#8.check subset and superset

```
a={4,5,9,8,7}
b={8,2,5,6,7,9,5,6}
print("subset:",a.issubset(b))
print("superset:",b.issuperset(a))"
```

""#9.find unique vowels in string

```
a="mileshmilan"
vowels=set()
for ch in a:
    if ch in "aeiou":
        vowels.add(ch)
print(vowels)""
```

""#10.convert list into set

```
a=[3,9,0,2,6,5]
b=set(a)
print(b)""
```

#dictionary tasks

""#1.create a student dictionary with name and marks

```
a={"name":"arun","marks":98}
```



```
print(a)"""
```

```
"""#2.add new key-value pair
```

```
a={"name":"arun","marks":98}
```

```
a["grade"]="A"
```

```
print(a)"""
```

```
"""#3.update dictionary value
```

```
a={"name":"arun","marks":98}
```

```
a["grade"]="A"
```

```
a["marks"]=90
```

```
print(a)"""
```

```
"""#4.delete an item from dictionary
```

```
a={"name":"arun","marks":98}
```

```
a["grade"]="A"
```

```
a["marks"]=90
```

```
del a["grade"]
```

```
print(a)"""
```

```
"""#5.print all keys and values
```

```
a={"name":"arun","marks":98}
```

```
a["grade"]="A"
```

```
a["marks"]=90
```

```
for key,value in a.items():
```

```
    print(key,":",value)"""
```

```
'''#6.search a key in dictionary
```

```
a={"name":"arun","marks":98}
```

```
a["grade"]="A"
```

```
a["marks"]=90
```

```
key="marks"
```

```
if key in a :
```

```
    print("key founded")
```

```
else:
```

```
    print("key not found")"""
```

```
'''#7.get all dictionary keys
```

```
a={"name":"arun","marks":98}
```

```
a["grade"]="A"
```

```
a["marks"]=90
```

```
print(a.keys())"""
```

```
'''#8.get all dictionary values
```

```
a={"name":"arun","marks":98}
a["grade"]="A"
a["marks"]=90
print(a.values())"
```

""#9.find length of dictionary

```
a={"name":"arun","marks":98}
a["grade"]="A"
a["marks"]=90

print(len(a))"
```

""#10.clear dictionary

```
a={"name":"arun","marks":98}
a["grade"]="A"
a["marks"]=90
a.clear()
print(a)"
```